

Research Statement: Quantum Machine Learning for IoT Intrusion Detection

This is the short version. The full write-up, with a related-work section and the complete discussion, is `docs/PAPER.md`; this document summarizes it and gives the roadmap.

1. Motivation

IoT deployments are projected to reach tens of billions of connected devices by 2030, a large share of that hardware resource-constrained: 8- to 32-bit microcontrollers, kilobytes of RAM, tight energy budgets. Classical intrusion detection typically requires substantial compute and large labeled datasets to reach 82-99% accuracy, a poor fit for this hardware. Quantum machine learning is proposed in the literature as an alternative: encode data into a small number of qubits, use superposition and entanglement for part of the representational work, and obtain comparable detection accuracy with fewer trainable parameters. This project tests whether that holds under a controlled, reproducible comparison.

2. Research question

Can a variational quantum classifier match classical ML baselines on binary IoT attack detection, and how does its performance scale with qubit encoding, ansatz depth, and qubit count? A secondary question: is there a point past which additional qubits or depth stop helping, and does that point align with known NISQ failure modes such as barren plateaus?

The result, from the experiment described in `docs/PAPER.md`: no, not at this scale. With sample size matched between the quantum and classical models, Logistic Regression and Random Forest both outperform the best variational quantum classifier tested, on a noiseless simulator capped at 8 qubits, and the gap does not close between 600 and 4,800 training rows. Circuit depth, not qubit count, determines the quantum model's own performance. This is a scoped simulator result, not a claim about quantum hardware or QML broadly; see the limitations below and in the paper.

3. Gap in the current literature

A review of 25 papers on QML for network and IoT intrusion detection, spanning 2020 to 2026 and independently checked for free full-text access, shaped the scope of this project. The consistent pattern: high headline accuracy, often above 99%, reported without a classical baseline trained on the same amount of data, and without results averaged across more than one random seed. Gouveia and Correia (2020), among the earliest papers in the area, state that their evaluation ran in simulation mode, which leaves open how the result would hold on hardware or under a different circuit design. Kalinin and Krundyshev (2022) and Abreu et al. (2024) both report quantum models outperforming classical ones on their respective datasets, but neither sweeps circuit depth, qubit count, or encoding systematically. Real-hardware validation of QML for this task turned out to be both rare in the literature and hard to find freely accessible: the one paper in this space with a matching four-architecture hardware comparison could not be confirmed as freely accessible and was excluded rather than cited without a working link (see `docs/PAPER.md`, Section 2, for the note).

This is not a criticism of that literature; a 99% accuracy figure on a well-defined attack-detection task is a legitimate result on its own terms. What has been comparatively rare is the specific comparison this project runs: same data, same sample size, multiple seeds, and a systematic test of which quantum-specific variable (encoding, depth, qubit count) accounts for performance. Two very recent papers, Gillani et al. (2026) and Cirillo, Esposito, and Seo (2026), run comparably controlled benchmarks on other datasets and reach the same conclusion (tuned classical models matching or beating quantum ones overall), so this project's result now joins that finding rather than standing alone against it. `docs/PAPER.md` covers all of this in more depth, with the full 25-paper reference list grouped by sub-topic.

4. Dataset

DataSense (also released as the CIC IIoT Dataset 2025), from the Canadian Institute for Cybersecurity at the University of New Brunswick (Firouzi, Dadkhah, Maret & Ghorbani), is a sensor- and network-synchronized benchmark for attack analysis in Industrial IoT. It provides matched network (.pcap) and sensor (.json) captures for benign and attack scenarios across a multi-device testbed, pre-aggregated into fixed time windows from 1 to 10 seconds, with per-window statistical features including packet size, TTL, TCP flags, and inter-arrival time.

This project uses the 3-second window: 71 numeric features after cleaning, approximately 75,000 labeled samples, 39% attack and 61% benign, with a binary target `IsAttack` built from the dataset's attack-family labels.

The raw processed CSVs are not redistributed here, since CIC datasets require agreeing to the institute's terms to obtain them, but a 2,000-row sample is committed so the pipeline can be run end to end without the full dataset. Full citation in §9.

5. Method, in brief

The pipeline (`src/prepare_data.py`, `src/train_baseline.py`) merges the benign and attack files, drops identifier and non-numeric columns, builds the binary target, and splits into train and test only afterward; scaling is fit on the training fold alone, inside an `sklearn.Pipeline`, so no information leaks from the test set into preprocessing.

Classical baselines (Logistic Regression, Random Forest) are evaluated the same way throughout: 5-fold stratified cross-validation on the training set, one held-out test pass at the end. The quantum classifier is built in PennyLane: angle encoding into a `StronglyEntanglingLayers` ansatz, with data re-uploading, trained by minimizing binary cross-entropy with Adam. An initial version without re-uploading failed a basic overfitting sanity check and was replaced; the full account is in `docs/PAPER.md`, Section 4.3. The main experiment sweeps qubit count (4, 6, 8) against re-uploading depth (1, 2, 3 layers), five seeds each, with classical baselines retrained on the same subsample at every qubit count so the comparison is fair, plus a scale-up test on the best configuration, from 600 up to 4,800 training rows.

6. Results, at a glance

Qubits	Best VQC (mean F1 $\pm$ std)	Matched Logistic Regression	Matched Random Forest
4	0.807 $\pm$ 0.013	0.772 $\pm$ 0.000	0.853 $\pm$ 0.002

6	0.822 ± 0.017 (best overall)	0.860 ± 0.000	0.861 ± 0.004
8	0.809 ± 0.015	0.863 ± 0.000	0.859 ± 0.001

Classical baselines are refit across the same 5 seeds as the VQC, so both sides report seed-to-seed variance on equal terms (Logistic Regression's ± 0.000 is the `lbfgs` solver's determinism, not a stability finding). At every qubit count, the matched classical models outperform the best VQC configuration: Random Forest by 4 to 5 points of F1 (Welch's t-test, $p < 0.01$ at every qubit count), Logistic Regression by a smaller and less consistent margin (the VQC outperforms it at 4 qubits). Circuit depth accounts for most of the variation in VQC performance: a single re-uploading layer is weaker and considerably less stable across seeds than two or three layers, while adding qubits at fixed depth changes little. Training the best configuration on up to 4,800 rows instead of 600 does not close the gap.

This reverses an earlier, single-run result obtained before the classical baseline was matched on sample size; that result is retained in the repository as part of the record, not presented as a finding. The full numbers, the earlier unmatched comparison, and the discussion of what this result does and does not establish are in `docs/PAPER.md`.

7. Roadmap

Phase	Status
0: Scoping, clean repository, dataset citation	Done
1: Classical baseline	Done
2: Working variational quantum classifier	Done
3: Depth/qubit-count sweep, matched sample size, multiple seeds	Done
4: Write-up	This document and <code>docs/PAPER.md</code>
5: Packaging and release	Repository structured, committed, and pushed

Remaining work: validating the best configuration on real quantum hardware (all results so far are simulator-only), testing other encodings such as amplitude or IQP embedding, and extending from binary detection to the dataset's multi-class attack labels. None of these are required to consider the current research question answered; they are the natural next steps.

8. Limitations

The classical baseline and the sweep both cover binary detection only; the dataset's richer multi-class labels are not used. Logistic Regression and Random Forest ran with close-to-default hyperparameters rather than a tuned search; a tuned search would widen, not narrow, the classical models' lead. The VQC's own hyperparameters (learning rate, optimizer, step count) were likewise fixed rather than tuned, a caveat that applies just as much to the quantum side of the comparison as to the classical side. The dataset comes from a single IIoT testbed, and results should not be assumed to transfer to a different device population or attack mix without further evaluation. The sweep tested one encoding, angle embedding, and capped training size at 4,800 rows to keep simulation time reasonable; the conclusion is scoped to that regime. The angle-encoding range is fit on the training fold and clips out-of-range test points to its boundary; this affects well under 1% of evaluated points at every qubit count (see `docs/PAPER.md`, Section 7). All experiments run on a noiseless simulator;

no claim is made about how the result would hold on real, noisy quantum hardware.

9. References

1. Firouzi, A., Dadkhah, S., Maret, S. A., & Ghorbani, A. A. (2025). *DataSense: A Real-Time Sensor-Based Benchmark Dataset for Attack Analysis in IIoT with Multi-Objective Feature Selection*. Canadian Institute for Cybersecurity, University of New Brunswick.
2. Gouveia, A. & Correia, M. (2020). *Towards Quantum-Enhanced Machine Learning for Network Intrusion Detection*.
3. Kalinin, M. & Krundyshev, V. (2022). *Security intrusion detection using quantum machine learning techniques*. Journal of Computer Virology and Hacking Techniques, 19, 125-136.
4. Abreu, D., Rothenberg, C. E., & Abelém, A. (2024). *QML-IDS: Quantum Machine Learning Intrusion Detection System*. arXiv:2410.16308.
5. Gillani, S. A., Baig, M. S. A., Shah, S. M., Ali, A., & Siddiqui, H. (2026). *How Quantum Is the Advantage? A Fair, Calibration- and Noise-Aware Benchmark and Attribution Audit of Quantum Machine Learning for Network Intrusion Detection*. arXiv:2608.18155.

The full 25-paper reference list this project drew on, grouped by sub-topic (DDoS-specific detection, general network/IoT detection, hardware-validated QML, and fair matched-baseline benchmarking), is in `docs/PAPER.md`, Section 2 and References.